

## RESEARCH ARTICLE

# On outdoor testing procedures of large samples of PV modules

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## Abstract

STC power control of PV module supply requires testing large samples of modules with low uncertainty. This paper analyses the feasibility of outdoor measurements with the modules kept at their operating positions. The classical procedure of recording *I-V* curves and translating them to STC in accordance with IEC-60891:2021 using the cell temperature directly observed at a few points of the rear of the module entails expanded uncertainties larger than 3% ( $k = 2$ ), which is too much for this procedure being accepted in quality controls with contractual consequences. A convenient procedure for overcoming this barrier consists in comparing the *I-V* curves of a tested and a reference module of the same type, both working under the same operating conditions. The latter is mostly secured if they are in adjacent positions. However, when the procedure is applied to large samples of PV modules kept in their operating positions, the distance between both modules can reach tens of meters and significant inter-module temperature differences can arise. A method for counterbalancing these differences consists of correcting the measured power values by considering the temperature difference observed at the back-sheet centres of the tested and the reference modules. That also provides clues to estimating the uncertainty of the results. This procedure has been applied in seven testing campaigns, carried out at commercial PV plants. Dedicated instrumentation, based on two radio linked *I-V* tracers, allowing the simultaneous measurement of the *I-V* curves and of the temperature at the centres of the reference and the tested modules, has been developed for that. The resulting uncertainties are similar to those corresponding to high-quality solar simulators and low enough for dealing, in practice, with strict quality control requirements.

## KEYWORDS

*I-V* curve, outdoor measurements, PV module, reference module, STC power, temperature dispersion, uncertainty

## 1 | INTRODUCTION

The supply of photovoltaic (PV) modules in PV plants is subject to strict quality controls to ensure their expected performance during the 30–35 years of their operating lifetime. That includes the periodic testing of the maximum power under Standard Test Conditions (STC),  $P_{\text{TEST}}^*$ , of a large sample of modules, which easily reaches several hundreds of specimens per plant. Uncertainties in the STC power results must be low enough for dealing with contractual acceptance/rejection criteria, usually about 2%.<sup>1</sup> The common belief that accurate PV measurements, as required by these quality controls, are easily done by solar simulators,<sup>2</sup> has led to the commercial development of mobile solar simulators for the indoor testing of PV modules at the PV plant sites.<sup>2–4</sup> Along similar lines, there is abundant literature on uncertainty in  $P_M^*$  tests for measurements with solar simulators<sup>5–9</sup> but rather far less for outdoor measurements. In fact, as far as we know, all the studies about uncertainty in outdoor measurements are devoted to the calibration of samples of just a few modules (e.g., golden modules) under rather strict operating conditions, even using temperature control devices (such as thermally insulated boxes) that help stabilizing the module temperature close to the STC values.<sup>10–13</sup>

However, testing with solar simulators or with temperature control devices usually obliges to remove and reinstall the tested PV modules from their operating positions, which is a cumbersome and expensive task and represents a risk for the physical integrity of the modules. It also entails logistic difficulties, thus limiting the sample size and, therefore, its representativeness. Hence, measuring outdoors and without manipulating the modules represents an appealing alternative. In fact, measuring a module this way only requires its terminals to be disconnected from neighbouring modules to allow an *I-V* tracer to be connected. Then, an *I-V* curve is recorded under real conditions: incident irradiance,  $G$ , and solar cell temperature,  $T_C$ . Finally, either the entire curve or just the  $P_M$  value is corrected to STC. This can be done very quickly but gives rise to the issue of the uncertainty in  $P_M^*$ .

This paper first reviews uncertainty in the measurement of  $P_M^*$  crystalline silicon PV modules, distinguishing between three independent contributions, respectively, derived from the  $G$  and  $T_C$  sensors calibration, from the Data Acquisition System (DAS) and from the procedures for measuring and correcting the experimental values to STC. The latter is where the essential difference between outdoor and indoor (solar simulator) testing lies and is the core of this paper. The more classic way of extrapolating experimental values to STC consists of correcting the entire *I-V* curve in accordance with IEC 60891,<sup>14</sup> using a  $T_C$  value observed in one or many points of the external rear side of the modules. Then, uncertainty derives from the choice of correction parameters, namely, the series resistance,  $R_s$ , and from  $T_C$  determination. In turn, the latter derives from both inter-cell (within a module),  $\Delta T_C^{\text{IC}}$ , and inter-module,  $\Delta T_C^{\text{IM}}$ , (within the same array) temperature dispersion mainly due to airflow turbulent and laminar regimes, respectively.<sup>15</sup> The interaction of wind with flat plates, such as PV generators and modules, entails transitions from laminar to turbulent regimes. The point at which transitions take place depends on wind speed and direction, as well as the orientation, tilt, and shape of the

plate. As a first idea, wind speeds,  $v_w$ , over 2 m/s cause turbulent regimes.<sup>16</sup> The characteristic instability of airflow turbulent regimes leads to question the representativeness of the points where  $T_C$  is measured. Besides, uncertainty in  $T_C$  derives from the thermal drop between the cell and the rear of the module,  $\Delta T_{\text{CR}}$ , which is also affected by airflow instability.

Another possibility, precisely the one chosen in this work, consists of directly deriving  $P_M^*$  from the relationship between the peak powers of a simultaneously measured tested and a reference module providing both work under the same operating conditions. This is what surely happens when modules are adjacent. However, with modules separated by a few meters or more, which is always the case when a large sample of modules is tested against a single reference, all kept at their own positions,  $\Delta T_C^{\text{IM}}$  can arise, causing the tested and the reference modules to work at different  $T_C$ . An original contribution of this paper is a method for counterbalancing the effect of such difference, which allows correcting the power results and provides clues for estimating uncertainty.

A second original contribution of this paper consists of the development of a dedicated instrumentation based on two radio linked *I-V* tracers, allowing the simultaneous measurement of the *I-V* curves and of the temperature at the centres of a reference and a tested module. Finally, this instrumentation has been employed in seven PV module testing campaigns carried out in different commercial PV plants. In the search for determining type A uncertainty, we have repeatedly measured corresponding samples of modules, up to a total of over 7000 *I-V* curves on 600 PV modules. We have been able to proceed at more than 500 measurements per day. Estimated uncertainties are slightly larger than those corresponding to high-quality solar simulators, but still low enough for dealing in practice with strict quality control requirements. Hence, the contribution of this paper helps to open a new door for outdoor testing of PV modules in commercial frameworks, representing an appealing alternative to mobile solar simulators.

## 2 | UNCERTAINTY IN PV MODULE CHARACTERIZATION

### 2.1 | Background

In general, the experimental determination of  $P_M^*$  proceeds through three successive steps: (1) exposing the module to a sun (real or simulated) together with  $G$  and  $T_C$  sensors, (2) recording its electrical response by means of an *I-V* tracer and simultaneously the  $G$  and  $T_C$  signals, and (3) translating this response (the entire *I-V* curve or just the maximum power) to STC. Each of these steps can be understood as a different uncertainty source, independent from each other. The contributions to uncertainty coming from these three steps deserve further comment.

In the following, we adhere to the analysis and terminology contained in the Guide to the Expression of Uncertainty in Measurements.<sup>17</sup> The standard uncertainties corresponding to these three

steps are denoted as  $u^{GT_C}$  (i.e., due to G and  $T_C$  sensors),  $u^{DAS}$  (i.e., due to the Data Acquisition System used for recording I-V curves and G and  $T_C$  signals), and  $u^{PROC}$  (i.e., due to the measurement and translation to STC procedures), respectively. The components of uncertainty are categorized as Type A,  $u_A$  (whose uncertainty is estimated as the standard deviation of a series of repeated determinations), and Type B,  $u_B$  (whose uncertainty is calculated from the instruments accuracy and calibration characteristics). In this work,  $u^{GT_C}$  and  $u^{DAS}$  are generally treated as Type B while  $u^{PROC}$  is estimated as a combination of components of both types. When testing a sample of  $N$  modules, it must be distinguished between the STC power of an individual module,  $P_i^*$ , and the average STC power of the whole sample,  $P_S^*$ . Corresponding uncertainties are denoted as  $u_{P_i}$  and  $u_{P_S}$ , respectively. Looking for clarity in presentation, we follow the general nomenclature:

$$u_{\text{Type, Result}}^{\text{Component}}$$

Depending on the context, the fields Type and Component can be elided. For example,  $u_{P_i}^{T_C}$  refers to the uncertainty component in  $P_i^*$  due to the uncertainty in  $T_C$ , and Type is not specified because this component, in turn, results from the combination of a Type A component, due to  $\Delta T_{CR}$ ,  $\Delta T_C^I$ , or  $\Delta T_C^{IM}$ , and a Type B component, due to the calibration of the  $T_C$  sensor and the acquisition channel. Along the same lines,  $u_{P_i}$  refers to the final uncertainty in  $P_i^*$ , and Component is not specified because it results from the combination of all the constituent components. Note that this nomenclature implies that a same uncertainty can be denoted in different, but still compatible, forms. For example, the final uncertainty in the calibration of the solar sensor,  $u_{CAL}$ , is denoted  $u_{A, P_i}^{CAL}$  when estimating the uncertainty component in the power of an individual module, due to that calibration.

Moreover, in this work, all the uncertainty components are considered mutually independent. Hence, the combined uncertainty,  $u$ , is given by

$$u = (u_A^2 + u_B^2)^{1/2} \quad (1)$$

Finally, it must be remembered that the expanded uncertainty for a given confidence interval is  $\pm ku$ , where  $k$  is the coverage factor. More often, expanded uncertainty is given for an approximate 95% confidence interval or, which is the same, for  $k = 2$ .

## 2.2 | Uncertainty derived from G and $T_C$ sensors

Uncertainty due to G sensor,  $u_G^G$ , mainly derives from both G sensor calibration and spectral mismatch between this sensor and the tested module. G can be measured by means of pyranometers or PV reference devices, either cells or modules. It is worth mentioning that the IEA recommendations for Qualification of PV plants<sup>2</sup> refer only to pyranometers (“[...] It is not recommended to use a pyranometer as an irradiance sensor. For these instruments, measurement errors are

caused by the long response time to irradiance fluctuation and the broad band spectral responsivity up to 4000 nm” page 39) and commercial irradiance sensors with c-Si solar cells (“[...] But also the use of c-Si irradiance sensors must be considered with care, because its spectral responsivity will not 100% match with the fielded PV modules, so that a spectral mismatch error cannot be fully eliminated”) but not to reference modules, which is precisely the option considered here. This oblivion is somehow surprising as reference modules are the option leading to minimum uncertainty. In fact, this is why they are specifically recommended for power measurements in the industry. For example, N. Taylor says: “The use of a full-size reference module is recommended in measuring other modules in order to achieve correspondence of dimensions, mechanical construction, optical properties ... so as to minimize discrepancies.”<sup>18</sup>

This is likely because the marketing of pyranometers and reference cells is today well developed, while the same is not true for full-size reference modules. Hence, consulting the available commercial offer of solar radiation sensors can lead to the feeling that reference modules are not standard commercial products in today's PV plants. In fact, it is easy to get them<sup>1</sup> by just selecting some specimens of the PV project's module supply and calibrating them in an independent laboratory after they have been stabilized against light-induced degradation (LID). The IES-UPM's experience in outdoor measurements for the power control of PV modules' supplies dates from 1993 (further details are given in Section 2.3.3) and is almost exclusively based on using reference modules. Reporting in detail about this experience (handling, inspection, primary and working references, recalibration periods, and so forth) is left for future works. Here, we will restrict ourselves to note that tested and reference modules, providing they are of the same batches, are matched in optical (spectrum and incidence angle modifier) and thermal properties, which avoids discrepancies in effective G and  $T_C$  calculation. Moreover, to avoid uncertainties due to different soiling affection, tested and reference modules must be cleaned before the test. Finally, in order to preserve the calibration, this methodology includes declaring one of the reference modules as a primary reference and keeping it in the dark at the site, while declaring the other modules as working references and using them for the tests.

Traceability chains for terrestrial c-Si PV devices' calibration are currently well established. A round-robin among nine different laboratories conducted within the European Metrology Research Programme concluded that the standard deviation of the different calibration values for the same reference cell was 0.81%.<sup>19</sup> An inter-comparison of high-efficiency c-Si module calibration between five reference Asian and European laboratories showed a short-circuit current agreement of the same order.<sup>20</sup> Calibration certificates of PV modules measured at STC from accredited laboratories typically declare that uncertainty in short-circuit current, which is closely the same as uncertainty in G when the modules are used as irradiance sensors, is about  $u_G^{CAL} = 1.8\%$ . However, IES-UPM experience includes comparisons of calibrations at CIEMAT (Madrid, Spain), TÜV (Shanghai, China), Fraunhofer-IES (Freiburg, Germany), and JCR (Ispra, Italy), resulting in rating disagreements well below  $\pm 1\%$ , which suggest

$u_G^{CAL} = 0.5\%$ . In fact, IES-UPM provides reference module calibration services with  $u_G^{CAL} \approx 1\%$ .<sup>13</sup> In summary,  $u_G^{CAL} = 0.5\% - 2\%$  can be considered a representative range of what can be get when paying attention to select a reliable and properly calibrated module.

It must be understood that the consideration of  $u_G^{CAL}$  can be avoided in some practical cases. When measuring the average degradation due to outdoor exposure through periodic tests of a relatively large sample of modules, the calibration uncertainty is duly compensated just by using the same sensor in all the tests, providing it is kept in the dark between the different tests to assure calibration stability. Another interesting case is the contractual imposition of a given calibration reference in the framework of a particular project. This is sometimes done in the control of the STC power of the module supplied for large-scale PV plants.<sup>1</sup> First, a set of reference modules is calibrated in a reference laboratory agreed upon between the manufacturer and the buyer. Then, this set of reference modules is distributed among the manufacturer and the independent laboratory in charge of the STC power control, to establish a common calibration for their respective solar simulators. Setting a common agreed reference module is precisely the procedure suggested in this work for avoiding calibration differences and uncertainties for the outdoor quality controls. That allows the consideration of the calibration uncertainty to be avoided when applying acceptance/rejection criteria to the batches of PV modules. Otherwise, calibration is considered as a Type B uncertainty component and contributes to the overall uncertainty budget (Equation 1).

Uncertainty due to  $T_C$  estimation,  $u_p^{T_C}$ , derives from both  $T_C$  sensor accuracy and from lack of representativeness of the measured value due to  $\Delta T_C^{IC}$ ,  $\Delta T_C^{IM}$ , and  $\Delta T_{CR}$ . Direct  $T_C$  measurements are typically made with PT1000 or IR sensors with an accuracy of about  $\pm 0.5^\circ\text{C} - 0.7^\circ\text{C}$ . According to the law of propagation of uncertainty and considering a power temperature coefficient of  $0.35\%/^\circ\text{C}$ , the corresponding uncertainty in power is  $u_p^{T_C\text{sensor}} = 0.25\%$ . For presentation reasons, uncertainties due to  $\Delta T_C^{IC}$ ,  $\Delta T_C^{IM}$ , and  $\Delta T_{CR}$  are treated later in this work together with the one corresponding to corrections to STC. However, it is also possible to derive  $T_C$  from the open circuit voltage,  $V_{OC}$ , of a reference module. The procedure is defined in the IEC 60904-5:2011 international standard.<sup>21</sup> This method intrinsically avoids the uncertainty derived from the sensor accuracy and from  $\Delta T_C^{IC}$  and  $\Delta T_{CR}$ . Because of that, this norm grants primacy to this method for determining the “true”  $T_C$  value. This is an additional reason for insisting on the superiority of reference modules as both  $G$  and  $T_C$  sensors for outdoor testing of PV modules. Other authors have brought to light the same idea.<sup>22</sup>

## 2.3 | Uncertainty derived from the Data Acquisition System

Uncertainty derived from the DAS depends on the quality of the instruments used for measuring the short-circuit current of the  $G$  sensor, for recording the  $I-V$  curve and for measuring the  $T_C$  signal. A suitable choice of good electrical standard equipment allows the

minimization of this uncertainty. For example, the  $I-V$  tracers developed in this work use calibrated resistors for measuring current with an accuracy of  $\pm 0.25\%$ , voltmeters with an accuracy of  $\pm 0.15\%$ , and data-loggers with 20 bits digitalisation. Calibration drift is minimized as an annual calibration is required for the equipment. The resulting combined uncertainty in power measurements, due to current sensors and electronic devices, is about  $u_p^{DAS} = 0.3\%$ .

## 2.4 | Uncertainty derived from the translation to STC and $T_C$ determination

### 2.4.1 | In accordance with IEC 60891

IES-UPM performed an experiment consisting of repeatedly measuring (more than 500 times along a 4-month period) a previously calibrated module. The obtained  $I-V$  curves were corrected to STC in accordance with IEC 60891, using the  $G$  value derived from the  $I_{SC}$  of the own  $I-V$  curves (once corrected in temperature) and two  $T_C$  values: the one measured by a PT1000 glued to the back centre of the module and the other derived from the  $V_{OC}$  of the own curves in accordance with IEC 60904-5. Regarding the test conditions, IEC 60904-1 standard<sup>23</sup> prescribes that, for referring measurements to STC,  $G \geq 800 \text{ W/m}^2$  and solar incident angle must be within  $\pm 5^\circ$ . Wind speed is not specifically addressed in IEC-60904-1, but it is restricted to less than 2 m/s in IEC-60891 for measuring the module temperature coefficients as a way of assuring laminar airflow and thereby thermal homogeneity and stability. Relevant lessons learned from this experience are<sup>24</sup>

- Respecting IEC 60904-1 conditions strongly limits the period allowed for measuring, which, in turns, limits the feasibility of outdoor testing in quality controls of PV plants. For example, limiting the incidence angle to  $\pm 5^\circ$  in static plants in Madrid restricts that period to less than 1 h per day and less than 3 months per year, and this gets worse when limiting  $v_w < 2 \text{ m/s}$ . Because of that, practical in-field testing campaigns in commercial frameworks must accommodate to larger operative ranges. Following with the case of static plants in Madrid, relaxing the validity conditions to  $G \geq 700 \text{ W/m}^2$  and  $v_w < 6 \text{ m/s}$  allows to more than doubling the available time for measurements. That means these measurements are often being performed under turbulent wind regimes. IES-UPM experiment has adhere to such more relaxing conditions.
- Errors in  $T_C$ , understood as the difference between the readings of the PT1000 and the values deduced from  $V_{OC}$ , are characterized by a standard deviation,  $\sigma = 1.5^\circ\text{C}$ , and a mean bias error,  $MBE = 1.3^\circ\text{C}$ . This suggests  $\Delta T_{CR}$  is quite variable, likely due to thermal transitions brought about by air turbulence and the variability of wind speed and direction. Equations for correcting  $\Delta T_{CR}$  have been proposed.<sup>25</sup> However, as they rely on the assumption of thermal stability, they allow correcting the error due to  $MBE$  but not that due to  $\sigma$ . According to the law of propagation of uncertainty and assuming a temperature coefficient of power of

0.35%/°C, the corresponding component of uncertainty in power can be estimated as  $u_{A,P_i}^{\Delta T_{CR}} \approx 0.53\%$  that, combined with the uncertainty due to the PT1000 accuracy,  $u_{B,P_i}^{PT1000} \approx 0.25\%$ , leads to  $u_{P_i}^{T_C} \approx 0.6\%$ .

IEC 60891 translation procedures require determining several correction factors. We followed the procedure termed as number 2, which requires five of these factors. Among them, the series resistance of the module,  $R_s$ , for which there are several possible choices, including available semi-empirical expressions,<sup>26–28</sup> information provided by the module manufacturer (PAN-files) and extraction from an experimental  $I$ - $V$  curve, as described in the own IEC 60891 standard. The point here is that selecting one or other entails differences in resulting power as large as  $\pm 2\%$ , which can be interpreted<sup>\*</sup> as  $u_{P_i}^{R_s} = 1.0\%$ . Along similar lines, the rest of correction factors add further degrees of uncertainty.

Note that, both together, measuring  $T_C$  at the back of the PV module and selecting a  $R_s$  value led to  $u_{P_i}^{T_C \& R_s} = 1.2\%$  and  $u_{P_i}^{T_C \& R_s} = 1.1\%$ . In other words, even disregarding  $u^{CAL}$  and  $u^{DAS}$  and assuming a perfect  $\Delta T_{CR}$  correction, the expanded uncertainty in  $P_i^*$  and  $P_s^*$  is, respectively, 2.4% and 2.2%. When considering more real circumstances ( $G$  measured by other device than a reference module and no perfect  $\Delta T_{CR}$ ), both grow well beyond 3%, which helps explaining why outdoor power testing based on observing  $T_C$  at the back of the module and correcting the  $I$ - $V$  curve in accordance with IEC 60891 is generally not accepted in contractual power controls of PV module supplies. To make things worse, a review of the operating manuals of commercial  $I$ - $V$  curve analysers has shown that the translation procedure in most cases does not follow IEC 60891 and the computing algorithm is in many cases not disclosed.<sup>2</sup>

## 2.4.2 | Comparison with reference modules

An appealing alternative consists of comparing the  $I$ - $V$  curves of the tested module and a reference module of the same type, both working under the same operating conditions. Then, the STC power of the tested module,  $P_{TEST}^*$ , is simply calculated as

$$P_{TEST}^* = P_{TEST} \frac{P_{REF}^*}{P_{REF}} \quad (2)$$

where  $P_{TEST}$  is the measured power of the module to be characterized and  $P_{REF}^*$  and  $P_{REF}$  are, respectively, the calibrated STC power and the measured power of the reference module. This is a relative measurement; that is, the result of the measurement,  $P_{TEST}^*$ , is established in relation with another measurement,  $P_{REF}$ . Note this procedure does not require neither estimating  $G$  and  $T_C$  nor using a parametric correction, which essentially nullifies the uncertainty components  $u_{P_i}^G$ ,  $u_{P_i}^{T_C \& R_s}$ , and  $u_{P_i}^{\Delta T_{CR}}$ . This is achieved at the price of handling with a somewhat

cumbersome reference module (typically, 2–3.5 m<sup>2</sup> and 35 kg), which is affordable in the context of a large-scale PV projects. Besides, this entails the additional advantage of allowing a direct link with the  $P_M^*$  measurements taken from solar simulators, as a part of in-factory quality controls of the PV modules supplied, just using a specimen of the set of reference modules distributed at the beginning of the project among the manufacturer and the independent laboratory in charge of that controls. As mentioned before, if this primary reference module is agreed between the parties, used for the PV modules' quality control and kept in the dark, calibration and calibration drift uncertainties can also be avoided.<sup>†</sup>

This procedure was originally developed and successfully applied in 1993 for controlling the module's supply of the 1 MW Toledo-PV plant, the first European project reaching the MW scale.<sup>29</sup> The selected modules were tested before their installation in the arrays, using a dedicated supporting structure able of carrying two tested modules plus the reference one. This way, the physical distance between the modules was kept below 2 m, which was considered enough for assuring that the modules operated at the same conditions.

This has been the procedure generally adopted by IES-UPM in many STC power quality controls for 30 years. We were able to proceed at about 20 tested modules per day, which was enough for the relatively low entity of the PV systems of this early period, typically below 1 MW. The steadiness of the procedure was analysed through the periodical measurements of three modules (10 times over 6 years).<sup>30</sup> Corresponding  $P_M^*$  results kept within a  $\pm 1\%$  around the mean value. Assuming this observed range corresponds to the 95% confidence level,  $u^{PROC} \approx 0.5\%$ . As reference, the difference between the  $P_M^*$  values obtained with two A<sup>+</sup>A<sup>+</sup>A<sup>+</sup> solar simulators (the one of the manufacturer and the one of an accredited independent laboratory), both calibrated with the same set of reference modules, on a sample of 4000 modules measured as part of the control of the STC power of a module supply of a 240 MW plant, were characterized by a mean bias error, MBE = 0.1%, and a standard deviation  $\sigma = 0.47\%$ .<sup>1</sup>

However, practical testing samples of the order of several hundred of modules requires much higher testing rates, which, in turns, obliges to keep the modules in their positions once installed in the arrays. Now, the tested and the reference modules are not adjacent but some tens of meters apart, and one wonders to what extent they are still working under the same operating conditions. In 2017, we applied this procedure (Equation 2) to some modules kept on their operating positions in a commercial PV plant. Three months before, these modules had been tested using a mobile solar simulator, which allowed us to compare our results with those from a solar simulator. The MBE of the differences between both methods was lower than 1%, which supported the validity of the outdoor method for relatively large samples of modules. However, we also observed that differences for individual modules reached up to 4% in the more extreme

\*That results from understanding the limits of the observed range as the upper and lower limits of a 95% confidence interval, so that, under a Gaussian hypothesis, their difference is equal to close four times the corresponding standard deviation.

†In case a project does not have a reference module agreed between the parties, or it has not been kept in the dark, the first step of the procedure is to create this reference module, by calibrating one of the modules of the supply. In case no modules of the same type will be available for creating the reference, an additional mismatch term must be added to the uncertainty calculation.

cases, in particular, when the tested and the reference modules were in places with different wind exposure, such as one in the centre and the other one on the edge of the array.<sup>31</sup> This agrees with findings from other authors<sup>15,32</sup> and suggest that the main reason for these differences is  $\Delta T_C^{IM}$ .

On the other hand, it is worth mentioning that this procedure allows minimizing other uncertainty sources, as those related to mounting, as PV generators' structures are already fixed and correctly aligned.<sup>‡</sup>

### 3 | A TWIST IN ACCURATE OUTDOOR TESTING

Encouraged by the challenges arising from the PV market growth, we decided to go one step further in the set-up of a testing procedure of commercial application, by incorporating two main innovations. On the one hand, looking for minimizing uncertainty, we attempted to counterbalance the errors due to  $\Delta T_C^{IM}$ , by estimating the temperature of the tested module as the temperature of the reference module (derived from  $V_{OC}$ ) corrected by the temperature difference observed between their respective centres. This leads to

$$\Delta T_{C,TR} = T_{C,TEST}^{Centre} - T_{C,REF}^{Centre} \quad (3)$$

$$T_{C,TEST} = T_{C,REF}^{V_{OC}} + \Delta T_{C,TR} \quad (4)$$

$$P_{TEST}^{*,C} = P_{TEST}^* \frac{1 + \gamma(T_{C,REF}^{V_{OC}} - T_C^*)}{1 + \gamma(T_{C,TEST} - T_C^*)} \quad (5)$$

where  $P_{TEST}^{*,C}$  is the corrected value of  $P_{TEST}^*$ . The corresponding absolute difference is denoted as  $\Delta P_i$ . So,

$$\Delta P_i = abs(P_{TEST}^* - P_{TEST}^{*,C}) \quad (6)$$

On the other hand, looking for performing at high testing rates, a decision was made in QPV, a start-up of IES-UPM devoted to technical consulting and testing services, to develop dedicated instrumentation consisting of two *I-V* tracers linked by radio and equipped with the means of measuring temperature. Then, one *I-V* tracer is permanently connected to the reference module and measures  $T_{C,REF}^{Centre}$  through a PT1000. The other *I-V* tracer moves from one test module to another and measures  $T_{C,TEST}^{Centre}$  using an IR sensor. Each time the *I-V* curve and corresponding  $T_{C,TEST}^{Centre}$  of a tested module is recorded (both of them obey the same order button), a radio signal is automatically

sent to the *I-V* tracer connected to the reference module, in such a way that both *I-V* curves are delayed by less than 20 ms. Then, the data corresponding to the reference module are automatically sent to the *I-V* tracer of the tested module, in which the whole set of data is processed. The *I-V* tracers are capacitive loads with four-terminal sensing. The internal capacitor is selected to ensure that the sweep time is larger than 40 ms. Once the *I-V* tracer is connected to the tested module, the whole measurement, data process and capacitor discharging, for allowing a new measurement, takes less than 5 s. That has allowed us to proceed to about 500 modules per day with only two technicians, providing the modules were previously cleaned and disconnected from the PV array, so that its terminals were easily accessible (see Figure 1). The conditions  $G \geq 700 \text{ W/m}^2$  and wind speed  $\leq 6 \text{ m/s}$  (as given by the meteorological station of the plant concerned) have been always respected.

A testing campaign of a sample of  $N_{MOD}$  modules at a PV plant is carried out through  $N_{ROUND}$  rounds. Each round of measurements consists of recording the *I-V* curve of each of the  $N_{MOD}$  modules simultaneously with the *I-V* curve of the reference module. Looking for independence between the thermal conditions of the different measurements for the same module, the consecutive rounds are delayed by a minimum of 2 h. The primary results of the campaign are, for each module,  $N_{ROUND}$  values of  $P_i^{*,C}$  and  $\Delta P_i$ . Assuming normal distribution, both are described by their corresponding means,  $\overline{P_i^{*,C}}$  and  $\overline{\Delta P_i}$ , and standard deviations,  $\sigma_{P_i}$  and  $\sigma_{\Delta P_i}$ , respectively. For the entire sample, that also leads to  $N_{ROUND}$  values of  $P_S^{*,C}$  and  $\Delta P_S$ , described, respectively, by  $\overline{P_S^{*,C}}$  and  $\overline{\Delta P_S}$  and  $\sigma_{P_S}$  and  $\sigma_{\Delta P_S}$ . Note that, by definition,



**FIGURE 1** Measuring process in one of the locations. The reference module is the one between the technicians, and is measured by one of the *I-V* tracers. The other tracer is connected one by one to the tested modules. [Colour figure can be viewed at [wileyonlinelibrary.com](http://wileyonlinelibrary.com)]

<sup>‡</sup>Selecting a representative position for the reference module becomes a key point in order to minimize bias due to misalignments. However, misalignments within a string (which is usually the measuring area when applying this procedure) are limited to very low values ( $\pm 0.5^\circ \text{C}$ ). The average impact of this maximum misalignments when measuring within  $\pm 2 \text{ h}$  from solar noon leads to  $u^{MISALIGN} < 0.24\%$ . This coefficient should be taken into account if a correct alignment cannot be assured.

$$\overline{P_S^{*,C}} = \frac{1}{N_{MOD}} \sum_i \overline{P_i^{*,C}} \text{ and } \overline{\Delta P_S} = \frac{1}{N_{MOD}} \sum_i \overline{\Delta P_i} \quad (7)$$

Besides, the  $\sigma_{P_i}$  values are also described by their corresponding mean and standard deviation values,  $\overline{\sigma_{P_i}}$  and  $\sigma_{\sigma_{\Delta P_i}}$ .

### 3.1 | Uncertainty estimation

In general, the standard uncertainty associated with the result of a measurement,  $u$ , can be expressed as

$$u = \sqrt{MBE^2 + \sigma^2} \quad (8)$$

where  $MBE$  and  $\sigma$  are, respectively, the mean bias error and the standard deviation of the frequency distribution of the differences between the “true value” and the results of a set of repeated observations of the same measurement. Based on the results of the testing campaign, we can make reasonable estimations of both values for two key parameters in power quality controls:  $\overline{P_i^{*,C}}$  and  $\overline{P_S^{*,C}}$ .

On the one hand,  $MBE$  is estimated as  $\overline{\Delta P_i}$  and  $\overline{\Delta P_S}$ , respectively. The rationale behind this is that, in the hypothetical case of the absence of thermal transients and a reference module located in a perfectly representative place, that difference will be null. As the ultimate source of uncertainty is the PV modules being at different thermal conditions and the selected results are corrected in temperature,

this  $MBE$  estimation is likely pessimistic; that is, it rather represents an upper limit than the most probable value. On the other hand,  $\sigma$  is estimated as  $\sigma_{\Delta P_S}$  and  $(\overline{\sigma_{P_i}} + 2\sigma_{\sigma_{\Delta P_i}})$ , respectively. Note that the last corresponds to an upper limit for 95% of the modules of the sample. Corresponding equations are

$$\mu_{P_S}^{PROC} = \sqrt{\overline{\Delta P_S}^2 + \sigma_{\Delta P_S}^2} \quad (9)$$

and

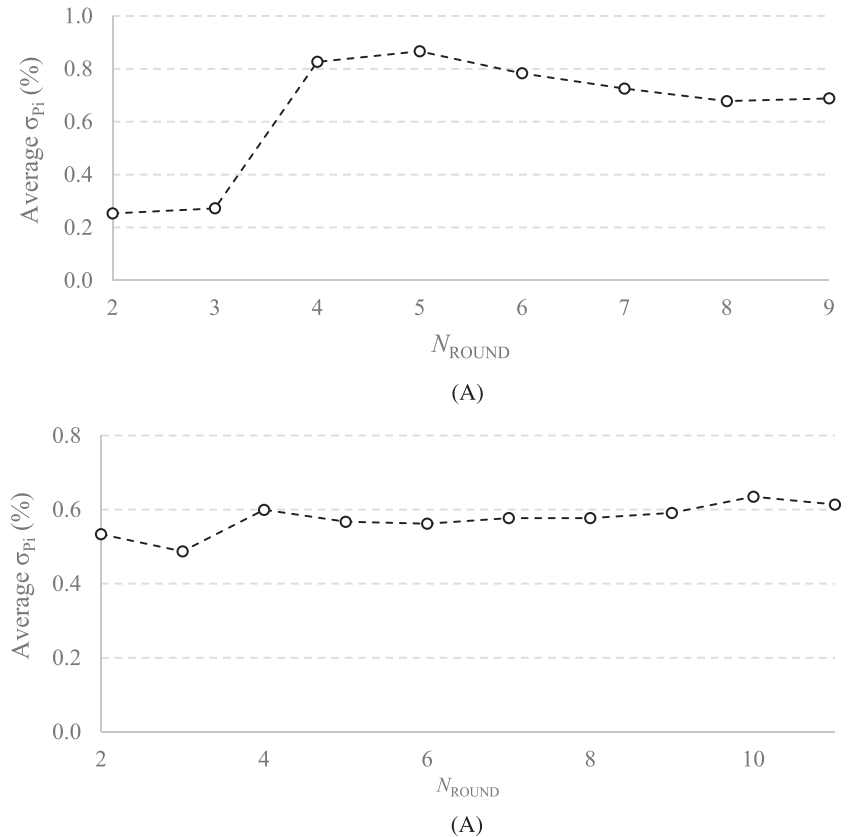
$$\mu_{P_i}^{PROC} = \sqrt{\overline{\Delta P_S}^2 + (\sigma_{P_i,95})^2} \quad (10)$$

where

$$\sigma_{P_i,95} = \overline{\sigma_{P_i}} + 2\sigma_{\sigma_{\Delta P_i}} \quad (11)$$

## 4 | TESTING CAMPAIGNS AT COMMERCIAL PV PLANTS

We have applied this methodology in seven testing campaigns in commercial PV plants located in Atacama and Coquimbo (Chile), Northern Cape (South Africa), Chihuahua and Sonora (Mexico), and Asuan



**FIGURE 2**  $\overline{\sigma_{P_i}}$  as a function of  $N_{ROUND}$  for the testing campaigns at (A) Sonora and (B) Atacama 1 PV plants.

(Egypt). First, we have explored the relation between  $N_{\text{ROUND}}$  and the procedure repetitiveness (characterized through  $\overline{\sigma_{P_i}}$ ) by measuring up to 11 rounds (see Figure 2). That leads us concluding that  $N_{\text{ROUND}} = 4-5$  represents a good compromise when thinking in the commercial application of this procedure. Beyond this number, repeatability hardly worsens.

We have been able to proceed at about 500 measurements per day. Table 1 presents the relevant information and results of the testing campaigns for  $N_{\text{ROUND}} = 4$ . Interestingly, the resulting uncertainties vary from one campaign to another, which must reflect different wind conditions and consequent different thermal transients in the PV arrays.

As regards the  $u_{P_s}^{\text{PROC}}$ , it is equal or less than 1% in six out of seven cases, which is good enough for most practical purposes (acceptance/rejection of PV modules supplies, evaluation of existing assets and even ageing). And the worst case,  $u_{P_s}^{\text{PROC}} = 1.08$ , is automatically and easily detected, so that the testing campaign can be repeated.

$u_{P_i,95}^{\text{PROC}}$  is about two times larger than  $u_{P_s}^{\text{PROC}}$ , but still useful for detecting defective or underrated modules (it is below 1.5% in five out of seven cases). These results lead us to believe that the procedure is useful for commercial applications.

On the other hand, we have seized the opportunity given by two of the testing campaigns to compare our results to those coming from an accredited testing lab that has measured the modules in the field with a mobile flash-tester. The total sample has been 229 modules. The mean absolute difference between both testing procedures for the average power of the sample is 0.71% while the standard deviation of the differences is 0.58% (see Table 2). The former is attributable to the inconsistencies between the different reference modules for each procedure (even being traceable to the same calibration origin), while the latter (which is the more relevant parameter here) is related to the stability and uncertainty of the procedure. This value is lower than  $\mu_{P_s}^{\text{PROC}}$  (0.71%) what can be understood as a further confirmation of the procedure accuracy.

**TABLE 1** Relevant information and results of the testing campaigns performed at seven commercial PV plants.

| PV plant                                                              | Coquimbo                                                                    | Kalahari                                                                     | Chihuahua                                                                   | Sonora                                                                       | Asuan                                                                        | Atacama 1                                                                    | Atacama 2                                                                    |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| PV module specifications                                              | Si-poli<br>$P_M^* = 315 \text{ W}$ ;<br>$\gamma = -0.41\%/^{\circ}\text{C}$ | Si-multi<br>$P_M^* = 295 \text{ W}$ ;<br>$\gamma = -0.45\%/^{\circ}\text{C}$ | Si-mono<br>$P_M^* = 320 \text{ W}$ ;<br>$\gamma = -0.40\%/^{\circ}\text{C}$ | Si-multi<br>$P_M^* = 295 \text{ W}$ ;<br>$\gamma = -0.45\%/^{\circ}\text{C}$ | Si-multi<br>$P_M^* = 295 \text{ W}$ ;<br>$\gamma = -0.45\%/^{\circ}\text{C}$ | Si-multi<br>$P_M^* = 295 \text{ W}$ ;<br>$\gamma = -0.45\%/^{\circ}\text{C}$ | Si-multi<br>$P_M^* = 295 \text{ W}$ ;<br>$\gamma = -0.45\%/^{\circ}\text{C}$ |
| PV array                                                              | Static, 20° tilt                                                            | 1 h-axis multirow                                                            | 1 h-axis single row                                                         | 1 h-axis multirow                                                            | 1 h-axis multirow                                                            | 1 h-axis multirow                                                            | 1 h-axis single row                                                          |
| Module disposition                                                    | 8L                                                                          | 1P                                                                           | 1P                                                                          | 1P                                                                           | 1P                                                                           | 1P                                                                           | 1P                                                                           |
| Testing period                                                        | 12–13 Oct 2018                                                              | 24–28 Oct 2018                                                               | 23 Feb–1 Mar 2019                                                           | 11–31 Oct 2021                                                               | 18–31 Mar 2022                                                               | 19–21 Oct 2022                                                               | 17–28 Oct 2022                                                               |
| $N_{\text{MOD}}$                                                      | 60                                                                          | 100                                                                          | 57                                                                          | 89                                                                           | 58                                                                           | 119                                                                          | 115                                                                          |
| Comparison with a reference module (Equation 2)                       |                                                                             |                                                                              |                                                                             |                                                                              |                                                                              |                                                                              |                                                                              |
| $\overline{P_S}(\text{W})$                                            | 321.9                                                                       | 299.7                                                                        | 327.7                                                                       | 326.3                                                                        | 316.7                                                                        | 321.4                                                                        | 407.5                                                                        |
| $\sigma_{P_S}(\text{W})$ ( $\sigma_{P_S}(\%)$ )                       | 1.00 (0.31)                                                                 | 1.90 (0.63)                                                                  | 0.72 (0.22)                                                                 | 2.53 (0.77)                                                                  | 4.08 (1.29)                                                                  | 2.85 (0.89)                                                                  | 1.97 (0.48)                                                                  |
| $\overline{\sigma_{P_i}}(\text{W})$ ( $\overline{\sigma_{P_i}}(\%)$ ) | 2.98 (0.92)                                                                 | 2.43 (0.81)                                                                  | 1.32 (0.40)                                                                 | 1.28 (0.39)                                                                  | 0.93 (0.29)                                                                  | 1.68 (0.52)                                                                  | 2.28 (0.56)                                                                  |
| $\sigma_{\sigma_{P_i}}(\text{W})$ ( $\sigma_{\sigma_{P_i}}(\%)$ )     | 1.60 (0.50)                                                                 | 0.66 (0.22)                                                                  | 0.83 (0.25)                                                                 | 0.98 (0.30)                                                                  | 1.01 (0.32)                                                                  | 0.77 (0.24)                                                                  | 0.63 (0.15)                                                                  |
| Refining the comparison with a reference module (Equations 3 to 5)    |                                                                             |                                                                              |                                                                             |                                                                              |                                                                              |                                                                              |                                                                              |
| $\overline{P_S}(\text{W})$                                            | 321.7                                                                       | 296.5                                                                        | 327.2                                                                       | 325.3                                                                        | 318.3                                                                        | 323.1                                                                        | 408.6                                                                        |
| $\sigma_{P_S}(\text{W})$ ( $\sigma_{P_S}(\%)$ )                       | 0.84 (0.26)                                                                 | 2.07 (0.70)                                                                  | 0.73 (0.22)                                                                 | 2.35 (0.72)                                                                  | 3.85 (1.21)                                                                  | 2.53 (0.80)                                                                  | 1.76 (0.43)                                                                  |
| $\overline{\sigma_{P_i}}(\text{W})$ ( $\overline{\sigma_{P_i}}(\%)$ ) | 2.34 (0.73)                                                                 | 2.81 (0.65)                                                                  | 1.66 (0.51)                                                                 | 2.65 (0.83)                                                                  | 1.03 (0.32)                                                                  | 1.94 (0.60)                                                                  | 4.01 (0.98)                                                                  |
| $\sigma_{\sigma_{P_i}}(\text{W})$ ( $\sigma_{\sigma_{P_i}}(\%)$ )     | 0.86 (0.27)                                                                 | 0.86 (0.28)                                                                  | 1.06 (0.32)                                                                 | 0.79 (0.24)                                                                  | 0.52 (0.16)                                                                  | 0.76 (0.26)                                                                  | 0.43 (0.11)                                                                  |
| Uncertainty estimation                                                |                                                                             |                                                                              |                                                                             |                                                                              |                                                                              |                                                                              |                                                                              |
| Sample average                                                        |                                                                             |                                                                              |                                                                             |                                                                              |                                                                              |                                                                              |                                                                              |
| $\Delta P_S(\%)$                                                      | 0.06                                                                        | 1.08                                                                         | −0.16                                                                       | 0.30                                                                         | 0.52                                                                         | 0.54                                                                         | 0.28                                                                         |
| $\sigma_{\Delta P_S}(\%)$                                             | 0.52                                                                        | 0.07                                                                         | 0.34                                                                        | 0.70                                                                         | 0.20                                                                         | 0.28                                                                         | 0.51                                                                         |
| $\mu_{P_s}^{\text{PROC}}(\%)$                                         | 0.52                                                                        | 1.08                                                                         | 0.38                                                                        | 0.76                                                                         | 0.56                                                                         | 0.61                                                                         | 0.58                                                                         |
| Single modules (upper limit for 95% of the sample)                    |                                                                             |                                                                              |                                                                             |                                                                              |                                                                              |                                                                              |                                                                              |
| $\mu_{P_i}^{\text{PROC}}(\%)$                                         | 0.83                                                                        | 1.42                                                                         | 0.58                                                                        | 0.94                                                                         | 0.71                                                                         | 0.90                                                                         | 0.67                                                                         |
| $\mu_{P_i,95}^{\text{PROC}}(\%)$                                      | 1.45                                                                        | 2.41                                                                         | 1.31                                                                        | 1.38                                                                         | 1.21                                                                         | 1.52                                                                         | 1.04                                                                         |

**TABLE 2** Comparison between the results obtained outdoors with the proposed methodology and those coming from an accredited testing lab with a mobile flash-tester in two of the locations.

| PV plant  | No. of modules | Measure type | $\overline{P}_S$ (W) | Diff $\overline{P}_S$ (W [%]) | $\sigma_{\text{Diff}-P_S}$ (W [%]) |
|-----------|----------------|--------------|----------------------|-------------------------------|------------------------------------|
| Atacama 1 | 115            | In-field     | 322.74               | −3.40 (−1.04)                 | 1.71 (0.53)                        |
|           |                | Flash        | 326.14               |                               |                                    |
| Atacama 2 | 114            | In-field     | 408.61               | 1.57 (0.39)                   | 2.56 (0.63)                        |
|           |                | Flash        | 407.07               |                               |                                    |
| Average   | 229            | -            | -                    | 2.49 (0.71)                   | 2.14 (0.58)                        |

## 5 | CONCLUSIONS

Outdoor testing of large samples of PV modules has been analysed, looking at the development of commercial controls of the STC power in module supplies, as an alternative to using mobile solar simulators. Outdoor testing allows the modules to be measured in their operating position, avoiding expensive and risky manipulation tasks. This is achieved at the price of having increased uncertainties due to errors in  $T_C$  coming from thermal transients and inhomogeneities, and due to the extrapolation process to STC. These errors give rise to a standard uncertainty component—additional to other components derived from the calibration of the irradiance sensor device and from the instrumentation for recording the  $I$ - $V$  curve—that we estimated as being larger than 1.2% and 1.1% for the power of an individual module and for the average of a large sample of modules, respectively, or, in other words, to expanded uncertainties ( $k=2$ ) larger than 2.4% and 2.2%. These rather high uncertainties, together with suspicions aroused by the selection of the  $R_S$  value and of the  $G$  sensor, help to explain why power testing based on observing  $T_C$  at the back of the module and correcting the  $I$ - $V$  curve in accordance with IEC 60891 is generally not accepted in contractual power controls of PV module supplies.

In this work, we have given preference to relative measurements, consisting of comparing the actual power of the module being tested with that corresponding to a reference module placed in a representative place on the PV array and whose power has already been agreed between the parties. This procedure requires neither estimating  $G$  and  $T_C$  nor using parametric corrections and entails the additional advantage of allowing a direct link with the  $P_M^*$  measurements taken from solar simulators, as a part of the in-factory quality controls of PV modules supplied. A methodology for counterbalancing the inter-module temperature dispersion, consisting of estimating the temperature of the tested module as the temperature of the reference module, derived from its  $V_{OC}$ , corrected by the temperature difference measured between their respective centres, has been proposed as a means of both improving accuracy and providing clues for estimating uncertainty.

Then, dedicated instrumentation based on two radio linked  $I$ - $V$  tracers, allowing the simultaneous measurements of a reference and a tested module, has been developed by QPV and employed in seven testing campaigns performed at different commercial PV plants. Each campaign consisted of measuring a sample of  $N_{MOD}$  modules being each module measured  $N_{ROUND}$  times. That instrumentation allowed

over 500 measurements to be taken per day, providing the modules were previously cleaned and disconnected from the PV array so that its terminals could be easily accessed. This represents more than twice the numbers currently achieved with mobile solar simulators. The results from these testing campaigns suggest that four testing rounds, delayed at least 2 h, allow obtaining average expanded uncertainties in STC power of about 1.3% for the average power of a sample of modules. This is similar to those corresponding to high-quality solar simulators and low enough for dealing in practice in the field with strict quality control requirements. Uncertainty in single module values is about twice larger but still suffices for detecting anomalous defective or underrated modules. The results have been validated against those obtained by an accredited lab with a mobile-flash, obtaining similar standard deviation values (<1%), what entails a further confirmation of the procedure validity.

In coherence with the strict conditions of typical contractual module power controls, all these testing campaigns have been performed with independently calibrated and properly preserved reference modules of the same brand as tested ones, and all the modules were properly cleaned before testing. One can imagine handling other practical cases, out of such strict conditions. For example, calibrating in-situ old exposed modules to create missing reference modules. Whichever the case, that would be at the price of increasing uncertainty.

## NOMENCLATURE

|                |                                                                                                                   |
|----------------|-------------------------------------------------------------------------------------------------------------------|
| DAS            | Data Acquisition System                                                                                           |
| $G$            | irradiance that reaches a PV module ( $\text{W}/\text{m}^2$ )                                                     |
| $N_{MOD}$      | number of modules measured during a testing campaign                                                              |
| $N_{ROUND}$    | number of rounds measured during the testing campaign                                                             |
| $P_M, P_M^*$   | maximum power (* stands for Standard Test Conditions) (W)                                                         |
| $P_{M,TEST}$   | measured maximum power of a tested module. Generic term (W)                                                       |
| $P_{M,REF}$    | measured maximum power of the reference module. Generic term (W)                                                  |
| $P_{M,REF}^*$  | calibrated power of the reference module. Generic term (W)                                                        |
| $P_{M,TEST}^*$ | maximum power of a tested module at STC. Generic term (W)                                                         |
| $P_i$          | power of an individual PV module. It may be accompanied by other subscripts and superscripts, as described above. |

For example,  $P_{i,TEST}^*$  describes the STC power of an individual tested module (W).

|                       |                                                                                                                                                                                                              |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $P_S$                 | power of the whole sample of PV modules. It may be accompanied by other subscripts and superscripts described above. For example, $P_{S,TEST}^*$ describes the STC power of the sample of tested modules (W) |
| $R_S$                 | series resistance of the PV module ( $\Omega$ )                                                                                                                                                              |
| STC                   | Standard Test Conditions                                                                                                                                                                                     |
| $T_C$                 | operating cell temperature of a PV module ( $^{\circ}\text{C}$ )                                                                                                                                             |
| $T_C^*$               | $T_C$ of a PV module at STC, that is, $25^{\circ}\text{C}$ ( $^{\circ}\text{C}$ )                                                                                                                            |
| $T_C^{VOC}$           | $T_C$ of a generic PV module, derived from its open-circuit voltage ( $^{\circ}\text{C}$ )                                                                                                                   |
| $T_{C,REF}^{VOC}$     | $T_C$ of the reference module, deduced from its open-circuit voltage ( $^{\circ}\text{C}$ )                                                                                                                  |
| $T_C^{Centre}$        | $T_C$ measured at the centre of a generic PV module ( $^{\circ}\text{C}$ )                                                                                                                                   |
| $T_{C,REF}^{Centre}$  | $T_C$ of the reference module, measured with a PT1000 sensor at its centre ( $^{\circ}\text{C}$ )                                                                                                            |
| $T_{C,TEST}^{Centre}$ | $T_C$ of the tested module, measured with a PT1000 sensor at its centre ( $^{\circ}\text{C}$ )                                                                                                               |
| $T_{C,TEST}$          | $T_C$ of the tested module, estimated from $T_{C,REF}^{VOC}$ and the temperature difference between the centre of the reference and tested modules ( $^{\circ}\text{C}$ )                                    |
| $V_{OC}$              | open-circuit voltage (V)                                                                                                                                                                                     |
| $V_{OC}^*$            | open-circuit voltage at STC (V)                                                                                                                                                                              |
| $V_{OC,REF}^*$        | calibrated open-circuit voltage of the reference module (V)                                                                                                                                                  |
| $v_w$                 | wind speed (m/s)                                                                                                                                                                                             |
| $\gamma$              | thermal coefficient of maximum power variation with temperature ( $\%/^{\circ}\text{C}$ )                                                                                                                    |
| $\Delta P_i$          | difference between $P_{i,TEST}^*$ corrected and not corrected by temperature differences between centres, for an individual PV module (W)                                                                    |
| $\Delta P_S$          | difference between $P_{S,TEST}^*$ corrected and not corrected by temperature differences between centres, for the whole sample (W)                                                                           |
| $\Delta T_C$          | difference between $T_C^{Centre}$ and $T_C^{VOC}$ ( $^{\circ}\text{C}$ )                                                                                                                                     |
| $\Delta T_C^{IC}$     | maximum inter-cell temperature difference measured inside a module ( $^{\circ}\text{C}$ )                                                                                                                    |
| $\Delta T_C^{IM}$     | maximum inter-module temperature difference measured inside an array ( $^{\circ}\text{C}$ )                                                                                                                  |
| $\Delta T_{CR}$       | difference between the temperature at the p-n junction of the cell and the one measured at the rear back-sheet of the PV module ( $^{\circ}\text{C}$ )                                                       |
| $\Delta T_{C,TR}$     | difference between temperatures measured at the centre of the tested and the reference modules, with PT1000 sensors ( $^{\circ}\text{C}$ )                                                                   |
| $u_p^{GT_C}$          | standard uncertainty in the calculation of $P_{TEST}^*$ due to the calibration of the irradiance and cell temperature sensors (%)                                                                            |
| $u_p^{DAS}$           | standard uncertainty in the calculation of $P_{TEST}^*$ due to the Data Acquisition System (%)                                                                                                               |
| $u_p^{PROC}$          | standard uncertainty in the calculation of $P_{TEST}^*$ due to the testing and extrapolation procedures (%)                                                                                                  |

|           |                                                                                  |
|-----------|----------------------------------------------------------------------------------|
| $u_p$     | standard uncertainty in the calculation of $P_{TEST}^*$ for an individual module |
| $u_{p_S}$ | standard uncertainty in the calculation of $P_{TEST}^*$ for a sample of modules  |

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## DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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